



Hub for Artificial Intelligence in Maternal, Sexual and Reproductive Health: HASH



Dr Elizabeth Oseku
Zindi Challenge Webinar
20th May 2026



What is HASH?

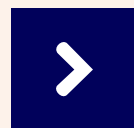
HASH is a multi-disciplinary consortium of 3 Ugandan partners:



Infectious Diseases Institute



Sunbird AI



Makerere University College of
Computing and Information
Sciences

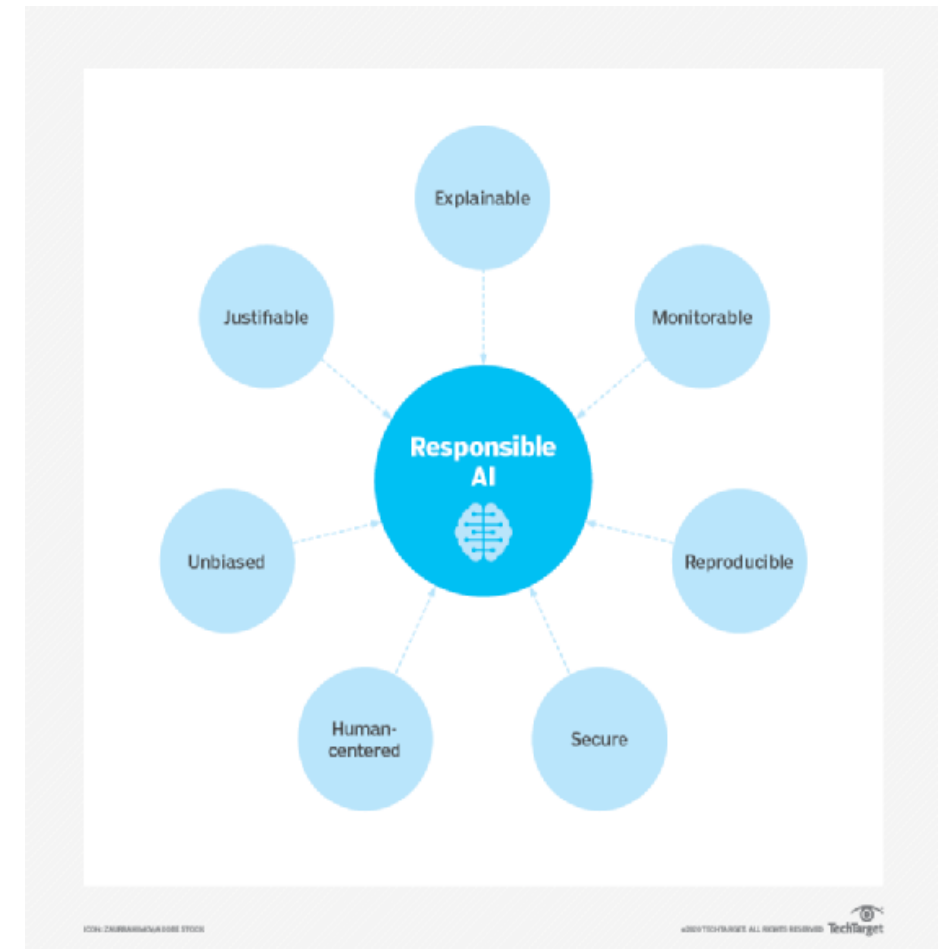
HASH is an African research Hub and Network for AI in Maternal, Sexual and Reproductive Health (MSRH) in Sub-Saharan Africa inceptioned in 2021.

HASH Principal Investigators



What do we do?

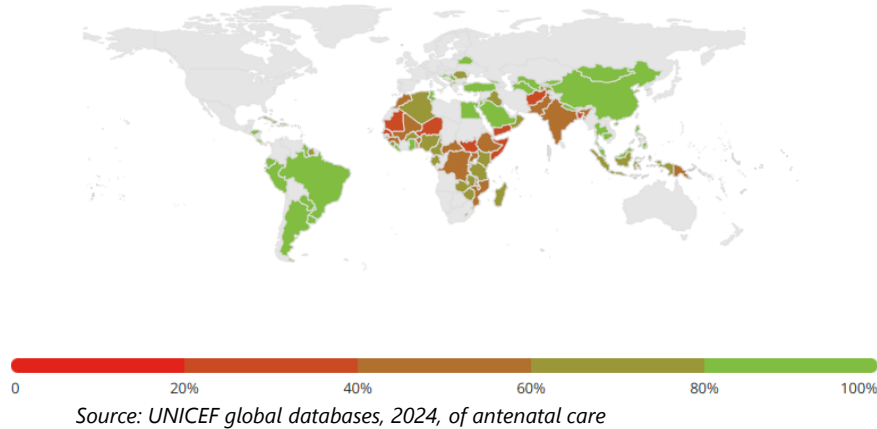
- **Goal:** To catalyse the development and scale of Responsible AI solutions that address maternal, sexual and reproductive health challenges in SSA.
- HASH targets pan-African anglophone, francophone and lusophone stakeholders working to enhance the use of AI and data technologies for MSRH.
- Grounded in the ethical and responsible application of AI, HASH provides funding and research capacity building through technical and methodological assistance to its members.
- HASH also provides opportunities for networking, collaboration and visibility.



Why Maternal, Sexual and Reproductive Health?



Percentage of women age 15-49 attended by any provider at least four times during pregnancy (ANC4) (2017-2023)

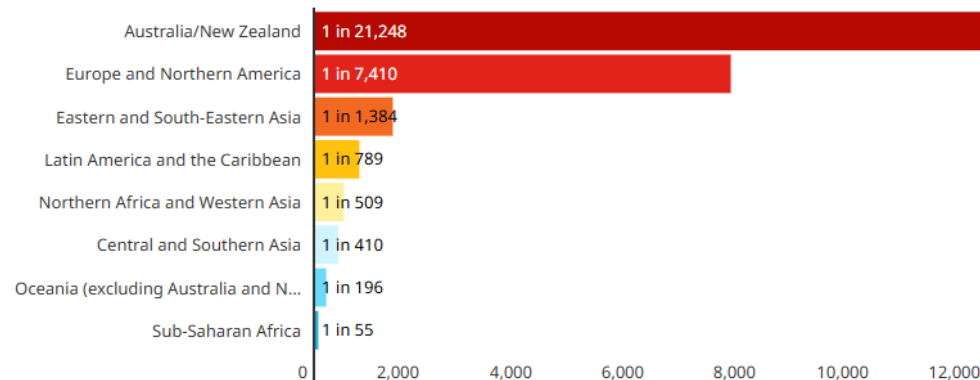


The highest rates of early childbearing are found in sub-Saharan African countries, where birth rates among adolescents in 2023 reached 93 births per 1,000 girls aged 15-19 years.

Source: UNICEF Data; Maternal Health Delivery Care, 2024

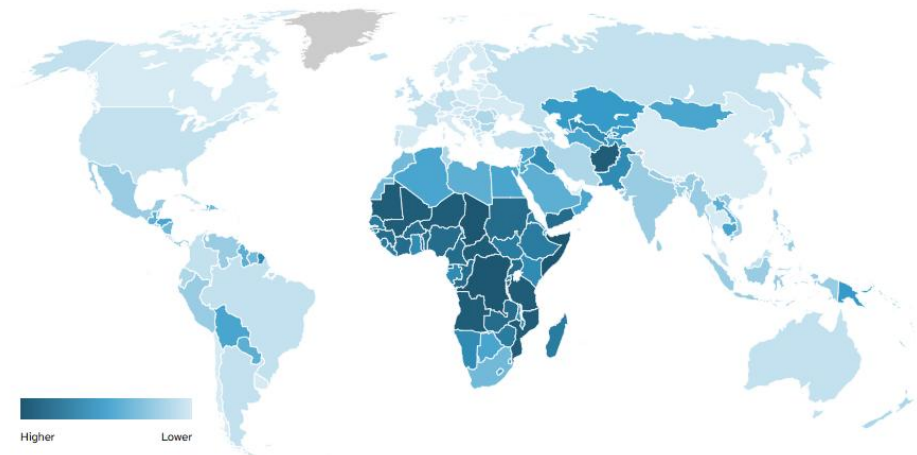
Lifetime risk of maternal death: 1 in X

By SDG region:



Source: WHO, UNICEF, and World Bank – Trends in Maternal Mortality 2000 to 2023, WHO Geneva, 2025

Total fertility rate, per woman, 2025

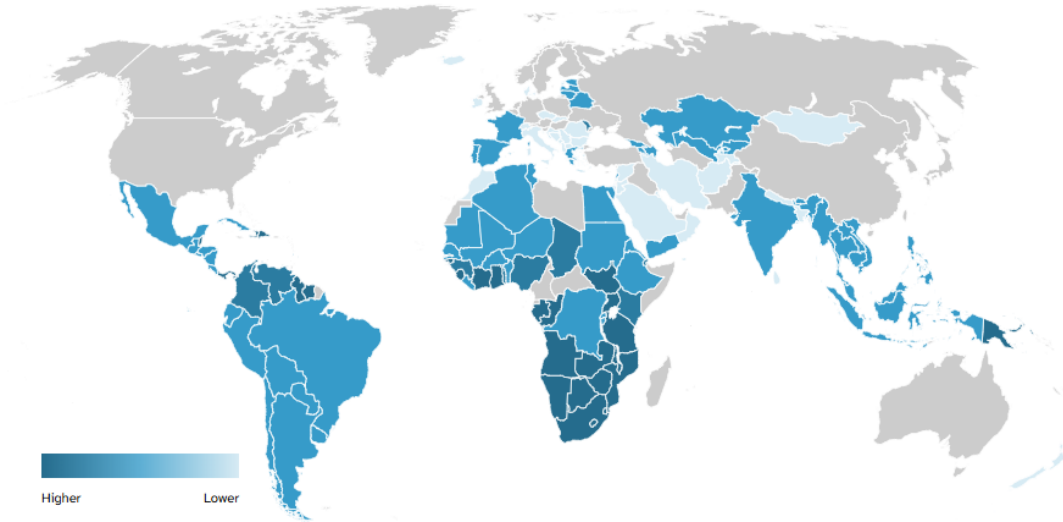


Source: UNFPA World Population Dashboard, 2025

Why Maternal, Sexual and Reproductive Health?

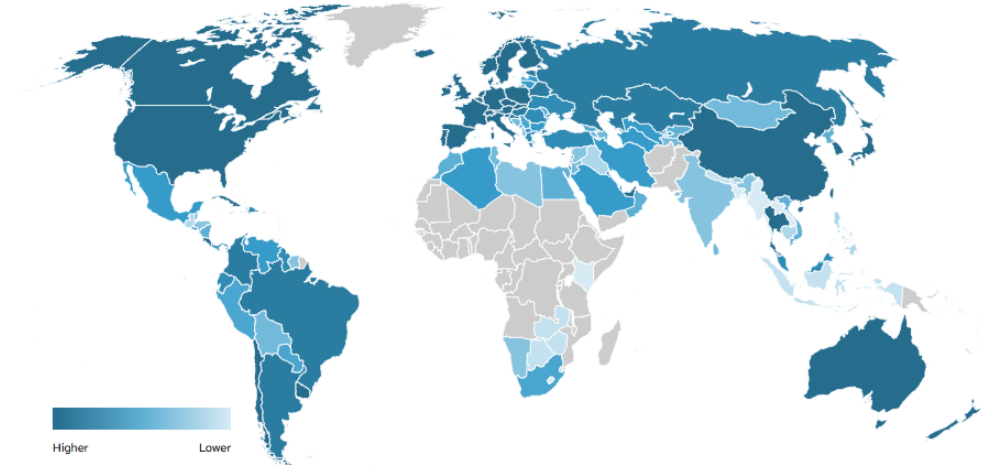


Number of new HIV infections, all ages, per 1,000 uninfected population, 2023



Source: UNFPA World Population Dashboard, 2025

Universal health coverage (UHC) service coverage index, 2021



Source: UNFPA World Population Dashboard, 2025

**Globally, more than 1 million STIs are acquired everyday. Most are in Africa.
Gonorrhea, HIV, Trichomoniasis treatments have rising levels of drug resistance.**

<https://www.who.int/teams/global-hiv-hepatitis-and-stis-programmes/stis/treatment/addressing-amr-in-stis>

Why Artificial Intelligence?

- Sub-Saharan Africa is experiencing a rapid expansion of digital connectivity, mobile technology, and data systems. Estimated 915M (53%) unique mobile phone users and 576M (33%) internet users by 2030.¹
- Digital health tools are increasingly being used innovatively for community health workers, telemedicine, and health records, etc.
- The African Union has adopted a Continental Artificial Intelligence (AI) Strategy guiding all 55 member states.²
- At least 15 countries have National AI Strategies.³
- Many countries are integrating AI into national development plans
- More than 36 African nations have passed data protection legislation.⁴
- Ordinary citizens are increasingly adopting AI. And By July 2025, 42.1% of Kenyan internet users aged 16 and older were using ChatGPT.⁵

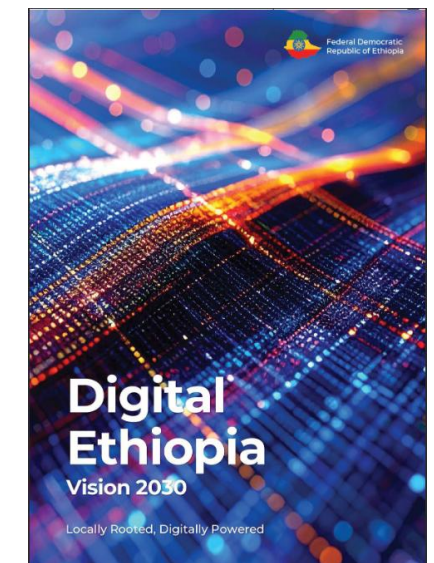
¹[GSMA Mobile Economy Report 2025](#)

²<https://au.int/en/pressreleases/20240617/african-ministers-adopt-landmark-continental-artificial-intelligence-strategy>, African Union 2024.

³ [ICT Legal Consulting, 2025](#)

⁴ [Data Protection Legal Regime and Data Governance in Africa: An Overview. Olumide Babalola, February 2022](#)

⁵ [South Africa Leads Africa's AI Adoption as Kenya, Nigeria and Others Accelerate Use.](#) iAfrica, January 2026



What is our contribution?



Research

Research Studies

- **HASH Qualitative study** – interviews members of the general population and expert stakeholders to explore perceptions on the opportunities, risks, limitations, and best practices for responsible AI applications in MSRH in Sub-Saharan Africa.
- **HASH Crowdsourcing Study** - Cross sectional survey enrolling English speaking participants to anonymously crowdsource questions on sexual health.

“the whole issue rotates around who is going to be the end user of the AI”



HASH Crowdsourced Dataset

- Questions on Sexually Transmitted Infections (STIs) and sexual health
- Two formats of answers (long and short)
- Total number of questions and answer pairs - 5,488
- Validated by health workers for accuracy
- Validated by community members for comprehension
- Published as an open source dataset on Harvard Dataverse.





Funding Innovators

HASH Phase I Subgrantees



mDoc Healthcare



KEM a health coach chatbot:

<https://mymdoc.com/complete-health>
<https://kem.mymdoc.com/>

Abstract presentation at GDHF “Empowering women through LLMs: Revolutionizing self-care in Low and Middle-Income Countries”

University of Ghana, Legon



Teenchat and #bot for adolescents with disabilities in Ghana

James Bumba



A machine learning tool for prediction of miscarriages among women seeking antenatal care in Uganda

The Medical Concierge Group



A Predictive tool for one's ability to pay for Pre-Exposure Prophylactic (PrEP) services in Uganda:

Bulk predictions: <https://proj-hash.streamlit.app/>
Individual Predictions: <https://hash-individual.streamlit.app/>

Makerere University



A machine learning-aided platform for point-of-care pregnancy risk assessment: Uterine

Fibroids classification model: <https://bit.ly/46l8CYL>

Uterine Fibroids detection and segmentation model: <https://bit.ly/4fgcl4s>

Mwavu Rodgers



AI model for predicting and recommending contraceptive options tailored to individual preferences



Addis Ababa Science and Technology University

A Sexually transmitted disease monitoring and assistance chatbot:

<https://mihret.pythonanywhere.com/#chatbot>

Rule based chatbot design methods: A review:

<https://doi.org/10.69660/jcsda.01012405>



University of Embu

Beshte Chatbot: <https://beshte.co.ke/>



Muhimbili University of Health and Allied Sciences

An AI- powered tool for TB screening among people living with HIV



Pan African Information Communication Technology

A machine learning-based predictive tool to identify at-risk patients for gestational hypertension.

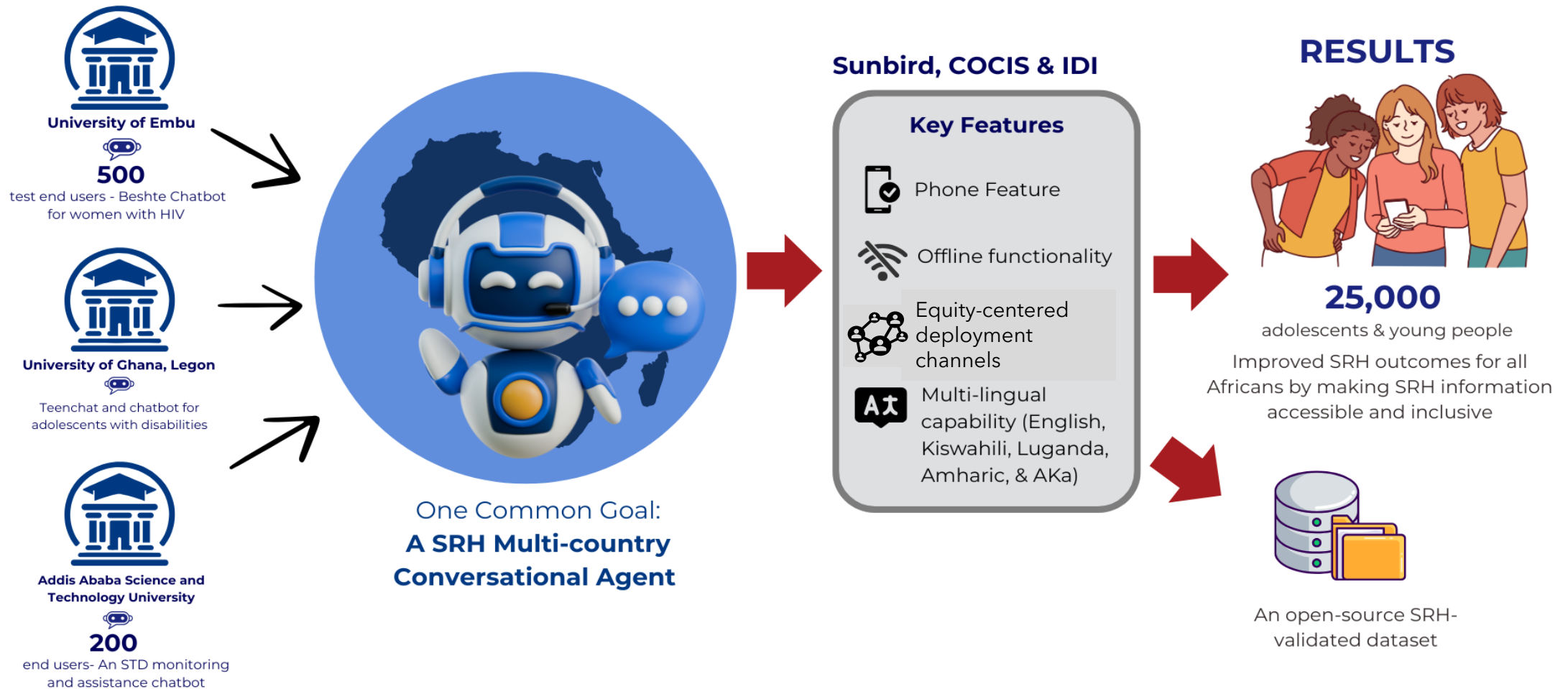
Project website: <https://paictahash.co.za/>

HASH Supported Innovations – Phase II



Co-creation of multi national conversational agent

Value Proposition: Making SRH information inclusive, accessible, usable and adaptive to address limited access to quality SRH information and increasing misinformation due to social media, traditional media, LLMs and peer





Capacity Building for Innovators

Innovation Challenges

- AI for Reproductive Health in Africa Innovation Challenge
- Zindi Challenge - Multilingual Health Question Answering in Low-Resource African Languages Challenge by ITU



AI FOR REPRODUCTIVE HEALTH IN AFRICA INNOVATION CHALLENGE

Join a unique innovation challenge, which empowers innovators to develop AI-powered solutions that tackle real-world healthcare challenges. Multidisciplinary teams from Sub-Saharan African countries will compete for an **award** and an all-expenses-paid trip to Uganda and attend **Data Science Africa** from 23rd - 24th July 2026.

Innovation Challenge Tracks

- 1 Early Pregnancy Loss Care
- 2 Empowering Informed Contraceptive Choices

Who is eligible to apply ?
Applicants must be residents or citizens of countries in Sub-Saharan Africa. Applicants from **Lusophone & Francophone** countries are encouraged to apply.

How to apply
Complete the eligibility form and submit a statement of interest.

Team Composition
Applications must be made by teams of **3-6 members** with at least 1 health worker & a member with knowledge or experience in machine learning, data science or software development. The team should have at least **2 women**.

Application Deadline
3rd May 2026

Apply here : <https://bit.ly/4clmKEG>

Scan this code to apply and be a part of this innovation challenge

REGISTER NOW ! REGISTER NOW ! REGISTER NOW ! REGISTER NOW ! REGISTER NOW ! REGISTER NOW !

Logos at the bottom: ARTIFICIAL INTELLIGENCE FOR DEVELOPMENT AFRICA, IDRC-CRDI, UK International Development, Sunbird AI, COLLEGE OF COMPUTING & INFORMATION SCIENCES, Academy for Health Innovation Uganda, DSA

Stakeholder Engagement

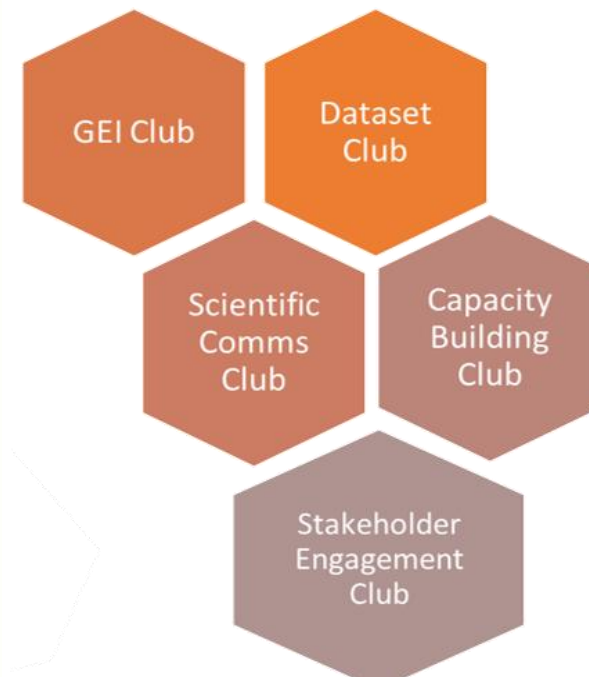
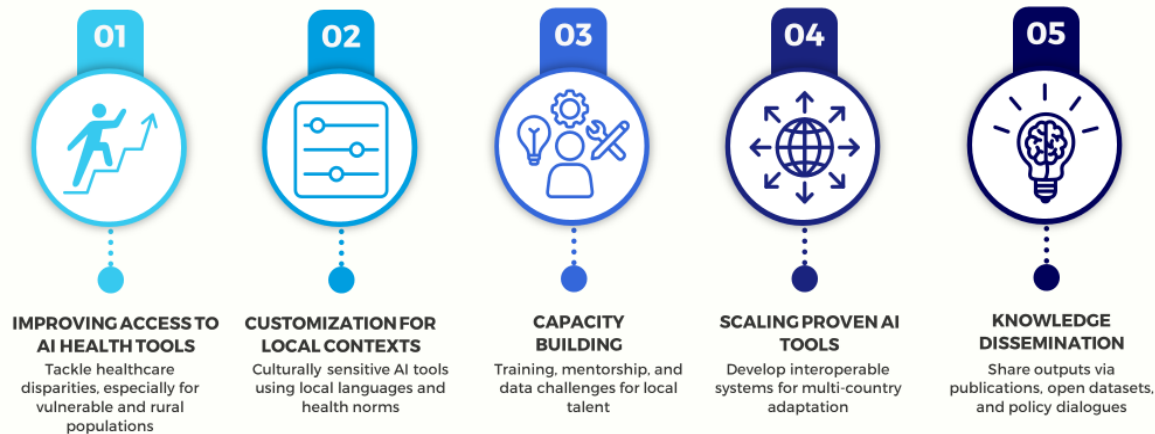
HASH Network

The HASH Network is our platform for collaboration, community and capacity building, launched in December 2022.

Value Proposition: “Creating a collaborative network for developing resilient and sustainable systems for Maternal Reproductive and Sexual Health (MSRH) through responsible AI”



HASH Network focus areas



How can you be part of HASH?

- Sign up to be a Network member
- Be part of the Zindi challenge
- Look out for updates from our ongoing hackathon in collaboration with APHRC

We would love to keep in touch!



Website: <https://hash.theacademy.co.ug/>

Email: hashproject@idi.co.ug

Twitter: <https://twitter.com/AiHub4MSRH>

LinkedIn: [The HASH Project](#)